

Candlelight - Test Data & Results

1. Testing Approach

Testing was conducted throughout development in accordance with the Quality Assurance Plan set out in Part 1, which specified functional, acceptance, live data, simulated data, beta and volume testing, supplemented by path and boundary testing on critical algorithms. Because my game was developed in a WAgile structure, testing was continuous rather than restricted to a singular phase. Each major feature has been tested against a structured test case before being marked as complete. A debug menu was added to track various in-game metrics to provide empirical evidence for my tests, including a visualiser for my enemy A* path-finding.

2. Functional Tests

Module	Input	Output
pixel_to_grid()	(464, 304)	(14, 9)
grid_to_pixel()	(14, 9)	(464, 304)
heuristic()	(2,1) and (5,3)	5
find_path()	Two connected floor tiles	Ordered list from start to goal
is_walkable()	Known floor tile	True
is_walkable()	Known wall tile	False
locate_candle()	count = 2	2 positions, each adjacent to 2+ walls
spawn_enemies()	count = 5	5 enemies spawned on floor tiles
get_current_spell()	slot = 2	(25, 40, 10, (255, 0, 0), False)
record_run()	score 50, rooms 3	profile updated in candlelight.db
get_profile()	played 2 rounds	returns 2 total runs, highscore

3. Boundary Testing

Boundary	Input	Output
Wax lower limit	wax = 0	Game Over
Wax just above zero	wax = 1, cast_cost = 10	Cast failed
Wax upper limit	wax = 95, +25 from relight	Wax caps at 100
Grid index below range	is_walkable(-1, 5)	False (no crash)
Grid index above range	is_walkable(30, 20)	False (no crash)
Grid corner	is_walkable(0, 0)	False
Spell queue wrap	Cast while slot = 3	Slot = 1
Enemy at exact range	Distance = enemy.range	Attack triggers
Enemy just beyond range	Distance = enemy.range + 1	No attack

4. Path Testing

4.1 Enemy AI

Branch	Conditions	Result
Out of range	Player far away	Pursues, no attack
In range, cooldown = 0	Player adjacent	Attack created
In range, cooldown > 0	Player adjacent	No attack
Horizontal	Player left/right	Enemy.direction = left/right
Vertical	Player above/below	Enemy.direction = up/down
No path available	Enemy sealed off	Enemy holds position

4.2 Cellular automata smoothing (*smooth_map()*)

Branch	Test grid	Result
Neighbours > 4	Tile with 6 wall neighbours	Becomes wall
Neighbours < 4	Tile with 2 wall neighbours	Becomes floor
Neighbours = 4	Tile with exactly 4	State unchanged
Border tile	Any edge tile	Remains wall

5. Performance and Volume Testing

Entity count	Uncapped FPS
2	500+
10	500+
20	500+
40	~450
100	~220
200	~150
300	~120